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Emissivity of silicon at elevated temperatures

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The temperature dependences of the spectral and total hemispherical emissivities of silicon have been experimentally determined, by using a technique which combines isothermal electron beam heating with *in situ* optical measurements. Emission spectra were used to deduce the absorption coefficient for phosphorus-doped silicon samples for wavelengths between 1.1 and 1.6 μm , in the temperature range from 330 to 800 $^{\circ}\text{C}$. For lightly doped samples, the data show good agreement with a model which includes the effects of the various phonon-assisted processes involved in interband transitions in silicon, as well as the free-carrier absorption. For heavily doped samples the agreement was less satisfactory, possibly because of inadequacies in the model for free-carrier absorption. It was shown that reflection spectra can also be used to determine the absorption spectrum, for the range where the absorption coefficient lies between 1 and $\sim 70 \text{ cm}^{-1}$. By fitting the theoretical model to the absorption coefficients derived from the reflection spectrum, it is possible to deduce the temperature of a sample, which is especially useful for temperatures less than 300 $^{\circ}\text{C}$, where the thermal emission is very weak. The total hemispherical emissivity of the specimens was determined from the input electron-beam power densities and the measured temperatures. The total emissivity of a 390- μm -thick specimen of lightly doped silicon rises from 0.12 at 280 $^{\circ}\text{C}$ to a limiting value of 0.7 at 650 $^{\circ}\text{C}$. This behavior is a consequence of the increase in the free-carrier concentration with the temperature. For heavily doped specimens the total emissivity remains approximately constant at ~ 0.7 between 200 and 800 $^{\circ}\text{C}$ because the carrier concentration is high even at room temperature, and the additional thermal generation of carriers produces an insignificant change in the total emissivity.

I. INTRODUCTION

The continuing development of rapid thermal processing (RTP) techniques for the fabrication of advanced electronic devices requires a detailed understanding of the processes of absorption and emission of radiation by semiconductors.¹⁻³ One of the main difficulties in thermal modeling of RTP is the lack of accurate data for the high-temperature optical and thermal properties of the materials being processed. For example, it is necessary to know the total hemispherical emissivity of a semiconductor wafer to calculate the equilibrium temperature it will reach when irradiated with a given power density, but its value is often not available in the literature. Furthermore, this quantity is usually a function of temperature and doping. In most RTP systems the wafer is heated by radiant energy emitted from lamps or resistively heated strips of graphite. If the heating power is supplied by tungsten-halogen lamps, then the absorption spectrum for wavelengths between 0.4 and 4 μm determines the efficiency with which the lamp energy is coupled to the wafer. For semiconductors this absorption spectrum can change very rapidly with temperature, as a result of the temperature dependence of the energy gap and the free-carrier concentration; however, the manner in which this takes place is often unknown. The effects of radiative heat loss and energy coupling are important in any situation where samples are not in thermal equilibrium with their surroundings, as is the case in RTP systems, molecular-beam-epitaxy (MBE) apparatus, and plasma-etching equipment.¹⁻⁶

Another area where the high-temperature optical

properties of semiconductors have a profound effect is in the use of radiation pyrometry for temperature measurement during RTP.^{1-3,7-9} Pyrometry is usually the only practical approach to the problem of determining the temperature of a wafer that is thermally isolated from its surroundings, but the technique is inaccurate if the emissivity of the sample is not accurately known. The use of the technique is also complicated by the fact that semiconductors can be partially transparent at the pyrometer wavelength, particularly if the wafer is lightly doped and at a low temperature. Similar problems are encountered when pyrometry is used in MBE systems.¹⁰

From these considerations it is evident that serious errors can arise in both theoretical modeling and experimental measurements if the optical properties of semiconductors are not known as a function of wavelength, temperature, and doping. In general, there is very little data available for the optical absorptivity and emissivity of semiconductors at elevated temperatures. Recently, a new technique which combines isothermal electron beam heating with *in situ* optical measurements was used to determine the temperature dependence of the total emissivity of GaAs.¹¹ This technique has now been extended to permit studies of the temperature dependence of the spectral, as well as the total, emissivity of semiconductors, and in this study it has been applied to the case of silicon.

To date there have only been a few studies of the high-temperature spectral emissivity of silicon. A recent study by Vandenabeele and Maex reported the temperature dependence of the emissivity of silicon wafers at wavelengths

of 1.7 and 3.4 μm .¹² Goldsmid *et al.* measured the spectral emissivity of very thin sheets of silicon in the spectral region between 0.9 and 2.4 μm , at temperatures between 760 and 920 °C.¹³ The most extensive study was by Sato, who measured the spectral emissivity of both heavily and lightly doped silicon samples over the temperature range from 270 to 800 °C, for the spectral region from 0.4 to 15 μm .¹⁴ For most of this range of experimental conditions the emissivity was measured directly by comparing the thermal radiation from the silicon to that from a blackbody at the same temperature. However, for wavelengths between 1.1 and 1.6 μm , the thermal emission is very weak at temperatures <500 °C, so the emissivity was deduced by the indirect method of measuring the reflectivity and transmissivity of the silicon. It is this part of the spectrum, which is where the absorption edge of silicon lies, which has been explored in detail in the work presented here. It has been possible to study this part of the spectrum, using samples that are only ~0.4 mm thick, because the background light level in the electron-beam heating system is very low, compared to that in a furnace.

Several researchers have used transmission measurements to determine the temperature dependence of the optical-absorption coefficient of silicon in the near infrared, but most of the detailed studies have been confined to temperatures below 200 °C, or have been at a few selected laser wavelengths.^{15–22} A number of models for the high-temperature optical properties of silicon have been developed on the basis of these measurements. Some theoretical models have also been developed to assist in the thermal modeling of the behavior of silicon samples in RTP systems.²³ The results reported in this article have been compared to theoretical predictions from a model developed in the light of these earlier studies.

II. EXPERIMENT

Since the apparatus used was very similar to that employed before,¹¹ only a brief description will be given here. The specimens were 10 mm square samples of <111>-oriented, double-side-polished silicon which were 390 μm thick. The high-resistivity samples had a resistivity of ~78 $\Omega\text{ cm}$ and the low-resistivity samples had a resistivity of ~0.012 $\Omega\text{ cm}$. These resistivities correspond to phosphorus doping levels of 5.5×10^{13} and $3.5 \times 10^{18}\text{ cm}^{-3}$, respectively.²⁴ The samples were supported in thermal isolation from their surroundings while they were irradiated from beneath by a rapidly raster scanned electron beam, which results in uniform isothermal heating. A two-color pyrometer (Ircon Modline R Series) viewed the undersides of the specimens through a viewport. Two sets of experiments were conducted on each type of specimen, one set in which emission spectra were recorded for a range of heating power densities, and the other in which reflection spectra were recorded.

In the case of the emission experiments, the light emitted in the direction of ~45° to the specimen normal passed through a viewport and was chopped at a frequency of 270 Hz before entering a monochromator which had its slits set to 1.5 mm width, giving it a bandwidth of 7.5 nm. The

radiation transmitted by the monochromator was detected by a sandwich construction detector, consisting of a 5-mm-diam Si photodiode mounted in front of a 3-mm-diam InGaAs photodiode. These detectors were connected in parallel and the combined signal was fed into a high-gain transresistance amplifier mounted within the same package as the detectors. The output from this detector passed to a lock-in amplifier and then to a microcomputer which was used for data storage. The wavelength range covered was from 0.96 to 1.57 μm and the spectral scanning speed was 3 nm/s, which allowed a spectrum to be recorded in ~200 s. The spectral response of the system was calibrated by heating the silicon sample to 800 °C as measured by the two-color pyrometer, and recording its emission spectrum. The response function was then obtained by dividing this spectrum by that calculated for a blackbody at 800 °C. No attempt was made to correct for the wavelength dependence of the emissivity of the silicon, because the variation in this quantity is very small over the wavelength range of interest, at 800 °C.¹⁴

For the reflection experiments, light from a quartz-halogen lamp was chopped at 270 Hz before entering the chamber where it was focused to a spot of ~1 mm diameter on the surface of the specimen. The arrangements for the collection and detection of the reflected light were the same as those described above, except that a double-side-polished GaAs wafer was placed between the exit viewport and the entrance slit of the monochromator. This wafer served as an order sorting filter, to prevent wavelengths below ~0.9 μm from reaching the detector. This precaution was necessary for the reflection spectra because the lamp produced a significant amount of radiation in the wavelength range from 0.48 to 0.79 μm , which could appear as a second-order spectrum at the exit slit of the monochromator. It was found that this was not a problem in the emission experiments, because the amount of radiation emitted by the sample at these wavelengths was negligible when compared to the first-order spectrum. For the reflection experiments, the spectral response of the system was calibrated by recording the reflection spectrum from a 0.5- μm -thick gold film evaporated onto a silicon substrate. In theory this should exhibit a reflectivity of ~0.98 over the spectral range of interest here.¹¹ The larger signal levels in the reflection experiments permitted the spectral scanning speed to be increased to 6 nm/s.

III. RESULTS AND DISCUSSION

A. The temperature dependence of the spectral emissivity of silicon

The emission spectrum is the product of the blackbody radiation spectrum corresponding to the temperature of the sample and its spectral emissivity. Figure 1 shows an example of the emission spectrum for a high-resistivity silicon sample. The spectrum displays a maximum at a wavelength that is close to that corresponding to the band gap of silicon at the temperature concerned. At wavelengths below this the specimen is opaque, and its spectral emissivity is determined by the reflectivity of the surface. At

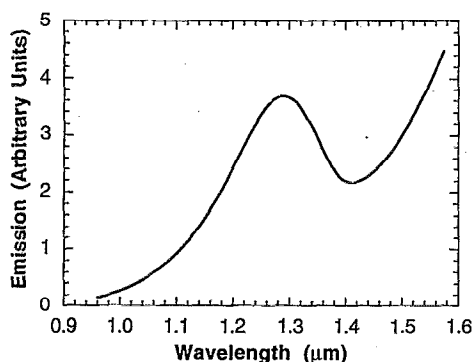


FIG. 1. The emission spectrum of a lightly doped silicon sample at a temperature of 501 °C.

longer wavelengths the specimen is semitransparent, and its spectral emissivity is determined by a combination of the reflectivity, the absorption coefficient, and the thickness of the wafer. The spectral emissivity is also a function of the angle of incidence and of polarization. If the refractive index $n \gg k$, where k is the extinction coefficient, then the emissivity is given by

$$\epsilon_{p(s)} = \frac{(1 - R_{p(s)}) [1 - \exp(-\alpha l)]}{[1 - R_{p(s)} \exp(-\alpha l)]}, \quad (1)$$

where $R_{p(s)}$ is the reflectivity of the interface for either the parallel (p) or the perpendicular (s) polarization, which will give two different values for the emissivity,²⁵ α is the absorption coefficient, and l is the effective path length through the sample, which is given by $l = d / \cos \theta$, where θ is the internal angle of incidence and d is the thickness of the sample. For unpolarized light the emissivity is given by

$$\epsilon_u = (\epsilon_p + \epsilon_s) / 2. \quad (2)$$

It should be noted that this expression does not in general give the same result as using

$$R_u = (R_p + R_s) / 2 \quad (3)$$

in an expression of the form of Eq. (1), instead of calculating ϵ_p and ϵ_s separately and using expression (2). In practice, the 45° angle of incidence used in this work does not result in a very large difference from the results expected at normal incidence. This is because silicon has a large refractive index, ~ 3.6 , which causes the internal angle of incidence θ to be rather small, so that $l \approx d$.²⁶ Another factor is that the value for R_u is 0.3204 if n is 3.6, which is almost the same as the value at normal incidence, 0.3195. Furthermore, it turns out that the error which arises in using R_u in an expression of the form of Eq. (1), instead of calculating ϵ_p and ϵ_s separately and using expression (2), is negligible under the conditions in this study.

The emission spectra of the samples were recorded for heating with a number of power densities. The two-color pyrometer has a minimum temperature readout of 700 °C, so the temperature of the samples was determined by fitting the Planck radiation function to that part of the spectrum for which the specimen is opaque. This corresponds

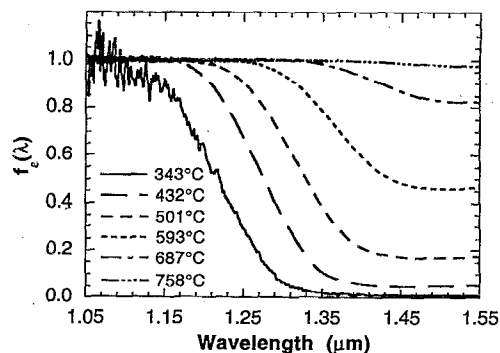


FIG. 2. The normalized spectral emissivity $f_e(\lambda)$ of lightly doped silicon for a range of temperatures.

to those wavelengths below the peak in the emission spectrum shown in Fig. 1. The procedure followed was the same as that described before, and it assumes that one can treat the silicon as a greybody over this wavelength and temperature range. This is a reasonable assumption, because the spectral emissivity is determined solely by the sample reflectivity if it is opaque, and the reflectivity of silicon displays a weak dependence on wavelength and temperature over the range of interest.²⁶

Once the temperature of the sample is known the spectral dependence of the emissivity can be obtained by dividing the emission spectrum by that calculated for a blackbody at an equal temperature. The results of such calculations are shown in Fig. 2 for a range of temperatures. The quantity $f_e(\lambda)$ is the spectral emissivity relative to that when the sample is opaque. Hence it equals 1 when the sample is opaque and 0 when the sample is transparent. The results shown in Fig. 2 illustrate the effect that the increase in temperature has on the spectral emissivity. At low temperatures $f_e(\lambda)$ decreases from 1 to 0 abruptly at the silicon absorption edge. As the temperature rises $f_e(\lambda)$ does not reach zero, but comes to a minimum before starting to rise slightly. This is the result of free-carrier absorption, which increases with wavelength and temperature. At temperatures above ~ 780 °C the free-carrier absorption is so large that the silicon is opaque over the entire wavelength range studied here.

The absorption coefficient α can be calculated from $f_e(\lambda)$ by using the expression

$$\alpha \approx \frac{1}{l} \ln \left(\frac{1 - f_e(\lambda) R_u}{1 - f_e(\lambda)} \right), \quad (4)$$

where the symbols have the meanings given above.¹¹ This expression is an approximation because it uses R_u instead of treating the R_p and R_s cases separately; however, under the conditions which apply in this study, its use results in negligible errors. The value of R_u used to calculate the results was 0.32, which is consistent with $n = 3.6$, which is a representative value for n , over the wavelength and temperature range considered here.²⁶ From the form of expression (4) it can be seen that the determination of α becomes prone to large errors as $f_e(\lambda)$ approaches 1. In the emission spectra noise has the most significant effect at short

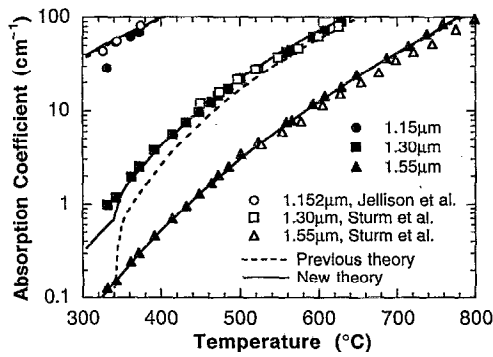


FIG. 3. The absorption coefficient of lightly doped silicon as a function of temperature. The solid symbols are the experimental data obtained in this work. "The experimental results of Jellison and Lowndes (see Ref. 19) and Sturm and Reaves (see Ref. 20)" are also shown. The dashed curves are theoretical predictions which do not take account of the transitions involving phonon energies corresponding to temperatures of 1050 and 1420 K. The solid curves are theoretical predictions from the new model presented in this work.

wavelengths and low temperatures, where the blackbody radiation is weakest. In this part of the spectrum it is also true that $f_{\epsilon}(\lambda)$ is often close to 1. This limits the range of temperatures and wavelengths over which expression (4) can be used to evaluate α .

Figure 3 shows α as a function of temperature at wavelengths of 1.15, 1.3, and 1.55 μm . These wavelengths are chosen so that the values of α presented here can be compared to those obtained by Jellison and Lowndes¹⁹ at a wavelength of 1.152 μm and by Sturm and Reaves,²⁰ whose studies were at 1.3 and 1.55 μm . The values of α obtained in this study are in good agreement with those presented in the previous studies. Sturm and Reaves explained their results by using a theoretical model for α which is similar to that given by Jellison and Lowndes, but they had to include the effect of free-carrier absorption, because their measurements were at longer wavelengths. Figure 3 includes predictions based on this theoretical model. The theory gives good agreement with the results for the wavelengths of 1.15 and 1.55 μm , but at 1.3 μm the agreement is only satisfactory for temperatures greater than 550 °C.

The analysis presented in the earlier studies was based on the work of Macfarlane *et al.*, who performed the first detailed examination of the band-edge absorption in silicon.¹⁵ In order to explain the results reported here, it is necessary to reexamine this original study and modify the previous models. The contribution to the absorption coefficient which arises from interband transitions α_{BG} is described by the relation

$$\alpha_{\text{BG}}(E, T) = \sum_{i=1}^4 [\alpha_{ia}(E, T) + \alpha_{ie}(E, T)], \quad (5)$$

where E is the photon energy, T is the absolute temperature, and $\alpha_{ia}(E, T)$ and $\alpha_{ie}(E, T)$ are the components associated with absorption and emission of the i th phonon, respectively. The sum is taken over the four types of phonon-assisted absorption process described by Macfarlane *et al.*¹⁵ and subsequently identified by McLean.¹⁶ The

four phonon energies involved were found to correspond to temperatures of 212, 670, 1050, and 1420 K. The energy and temperature dependences of the absorption coefficients for the processes involving the transverse acoustic ($i=1$) and optical ($i=2$) phonons are summarized by the expression

$$\begin{aligned} & \sum_{i=1}^2 [\alpha_{ia}(E, T) + \alpha_{ie}(E, T)] \\ &= \frac{1}{E} \sum_{i=1}^2 \left(\frac{F_i [E - E_g(T) + k\theta_i]}{[\exp(\theta_i/T) - 1]} \right. \\ & \quad \left. + \frac{F_i [E - E_g(T) - k\theta_i]}{[1 - \exp(-\theta_i/T)]} \right), \end{aligned} \quad (6)$$

where $E_g(T)$ is the temperature-dependent band gap, k is Boltzmann's constant, and the energies are in units of eV.¹⁵ θ_1 is 212 K and θ_2 is 670 K. The function $F_1(x)$ has units of eV cm^{-1} , and is of the form

$$F_1(x) = \begin{cases} 0.504\sqrt{x} + 392(x - 0.0055)^2 & \text{for } x \geq 0.0055, \\ 0.504\sqrt{x} & \text{for } 0 < x < 0.0055, \\ 0 & \text{for } x < 0. \end{cases} \quad (7)$$

The function $F_2(x)$ has units of eV cm^{-1} , and is of the form

$$F_2(x) = \begin{cases} 18.08\sqrt{x} + 5760(x - 0.0055)^2 & \text{for } x \geq 0.0055, \\ 18.08\sqrt{x} & \text{for } 0 < x < 0.0055, \\ 0 & \text{for } x < 0. \end{cases} \quad (8)$$

The expressions used for $F_1(x)$ and $F_2(x)$ are the same as those given by Jellison and Lowndes.¹⁹ They are an approximation of the original expressions given by Macfarlane *et al.*¹⁵ However, expression (6) includes the photon energy in the denominator, a factor that was omitted in previous models.

Macfarlane *et al.* also identified the components of the absorption coefficient which are associated with phonon absorption processes with characteristic energies equivalent to temperatures of $\theta_3=1050$ K and $\theta_4=1420$ K.¹⁵ Although they did not give expressions for the corresponding absorption coefficients, their data at 363 K are consistent with the expressions

$$\alpha_{3a}(E, T) = \begin{cases} \frac{536[E - E_g(T) + k\theta_3]^2}{E[\exp(\theta_3/T) - 1]} & \text{for } E \geq E_g(T) - k\theta_3, \\ 0 & \text{for } E < E_g(T) - k\theta_3, \end{cases} \quad (9)$$

and

$$\alpha_{4a}(E, T) = \begin{cases} \frac{988[E - E_g(T) + k\theta_4]^2}{E[\exp(\theta_4/T) - 1]} & \text{for } E \geq E_g(T) - k\theta_4, \\ 0 & \text{for } E < E_g(T) - k\theta_4, \end{cases} \quad (10)$$

which give values for α_{3a} and α_{4a} in units of cm^{-1} . They were unable to identify the components of the absorption coefficient associated with the corresponding phonon emis-

sion processes, because these are insignificant compared to other absorption processes; hence, it will be assumed that they can be ignored. McLean identified α_{3a} and α_{4a} as being associated with absorption processes which involve two phonons, and hence expressions (9) and (10) must be regarded as empirical approximations, because they are not derived from a full analysis of all the interactions involved. In particular, the temperature dependence might be expected to differ from that predicted by Eqs. (9) and (10) because of the different phonon energies involved. A more complete analysis would require consideration of all the combinations of two phonons which could be involved, which is beyond the scope of the current investigation.

The energy gap is calculated from the expression

$$E_g(T) = E_g(0) - AT^2/(B+T), \quad (11)$$

where $A = 4.73 \times 10^{-4}$ eV/K, $B = 635$ K, and $E_g(0)$ is taken to be 1.155 eV, as suggested by Jellison and Lowndes.¹⁹

The absorption coefficient resulting from free-carrier absorption α_{FC} was calculated using the formula given by Sturm and Reaves.²⁰ According to this model,

$$\alpha_{FC} = n(T)\sigma_n(T) + p(T)\sigma_p(T), \quad (12)$$

where $n(T)$ and $p(T)$ are the electron and hole concentrations, and $\sigma_n(T)$ and $\sigma_p(T)$ are the electron and hole absorption cross sections, respectively. The latter are given, in units of cm^2 , by

$$\sigma_n(T) = 1.01 \times 10^{-20} \lambda^2 T \quad (13)$$

and

$$\sigma_p(T) = 0.51 \times 10^{-20} \lambda^2 T, \quad (14)$$

respectively, where T is the temperature in K and λ is the wavelength in μm .²⁰ $n(T)$ and $p(T)$ are calculated using standard expressions for the temperature dependence of the concentration of electrons and holes in doped silicon.²⁴ The intrinsic carrier concentration was calculated from the data of Morin and Maita.²⁷

The total absorption coefficient is given by²⁰

$$\alpha = \alpha_{BG} + \alpha_{FC}. \quad (15)$$

Figure 3 shows that the predictions from this theory agree well with the experimental results, over the wavelength and temperature ranges covered in this work. Figure 4 shows a more complete comparison of the theory and the results at a larger number of wavelengths, and Fig. 5 shows a comparison with the experimental results for the wavelength dependence of α at a number of temperatures. For wavelengths longer than $1.45 \mu\text{m}$ the theory agrees with the results over the entire range of α which was measured. At these wavelengths α_{FC} is the major component of α over the temperature range covered in this work, which suggests that the model for α_{FC} provides an accurate description of the absorption produced by thermally generated free carriers, at least for this wavelength range. Recently, Vandenabeele and Maex presented an empirical model which can be used to predict the value of α_{FC} , for lightly doped silicon, for wavelengths between 1.5 and 5

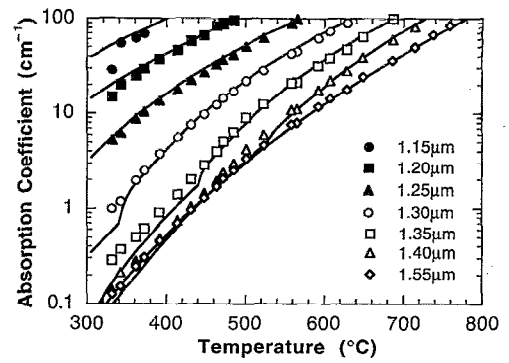


FIG. 4. The absorption coefficient of lightly doped silicon as a function of temperature. The points are experimental results for the wavelengths indicated. The solid curves are theoretical predictions from the model presented in this work.

μm and temperatures between 400 and 700 °C.¹² This model gives results that are the same as those reported here for wavelengths greater than $1.5 \mu\text{m}$.

The agreement between theory and experiment is also very good for wavelengths less than $1.3 \mu\text{m}$. In Fig. 4 the values for α at 330 °C, for 1.15 and $1.2 \mu\text{m}$, are lower than the theory predicts, but this is probably the consequence of experimental error, because the signal level at short wavelengths and low temperatures is very small. A more significant discrepancy between theory and experiment is evident in the results for wavelengths in the range $1.3 \mu\text{m} < \lambda < 1.45 \mu\text{m}$, for temperatures below 600 °C. For these conditions the experimentally determined values for α are higher than those predicted by the theory. Furthermore, a distinct “knee” is evident in the predicted absorption spectra shown in Fig. 5. A similar feature appears in the predictions of the temperature dependence of α in Fig. 4. This knee is associated with the onset of absorption α_{2a} , the component of α which is associated with the absorption of a transverse optical phonon. Macfarlane *et al.* found that absorption associated with this process occurred at energies below the threshold energy for this process, which is defined by¹⁵

$$E_{2a}(T) = E_g(T) - k\theta_2. \quad (16)$$

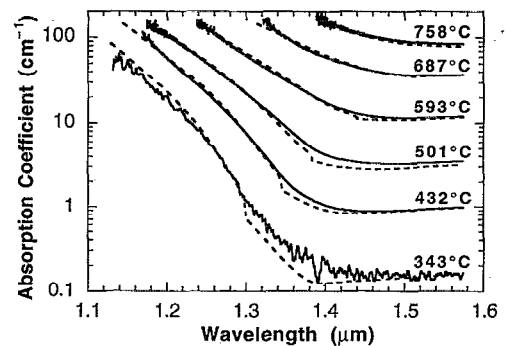


FIG. 5. The absorption spectra of lightly doped silicon at a range of temperatures. The solid curves are experimental results. The dashed curves are theoretical predictions from the model presented in this work.

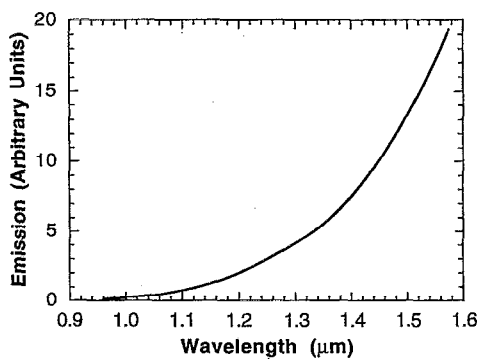


FIG. 6. The emission spectrum of a heavily doped silicon sample at a temperature of 504 °C.

This subthreshold absorption arises because of thermal broadening of the basic absorption shape defined by expression (8). The thermal broadening effects are related to the relaxation processes suffered by the particles involved in the optical-absorption process. It is possible to derive the absorption shapes at finite temperatures by applying a Gaussian type of broadening to the basic shape, but this procedure has not been used in the model used in this work because there is insufficient data available in the literature to provide a reasonable estimate of the relaxation time for temperatures above 150 °C.¹⁵ Since the broadening effects, which make $\alpha_{2a} > 0$, for $E < E_{2a}(T)$, are not included, the theory presented here underestimates α , as can be seen in Figs. 4 and 5. Other thermal broadening effects, associated with the other components of α , are also likely to be involved. More detailed studies of these results may help to clarify the situation, but the analysis is likely to be complicated by the fact that these phenomena occur at rather low absorption levels, where $\alpha_{BG} < 2 \text{ cm}^{-1}$, and α_{FC} is similar in magnitude. Analysis of α_{BG} would require the subtraction of α_{FC} from α , which would greatly magnify the effect of experimental errors and noise. The interpretation of the results would also be strongly affected by the accuracy of the model for α_{FC} .

A similar set of experiments was conducted on the heavily doped silicon. In this case the shape of the emission spectrum is rather different, because the doping results in a high degree of absorption in the silicon beyond the absorption edge, even at room temperature. A typical emission spectrum is shown in Fig. 6. The procedure used to determine the temperature was similar to that applied in the case of the lightly doped silicon. In some experiments, where the temperature reached was less than 300 °C, the temperature value was deduced by fitting the blackbody radiation function to the whole spectrum, rather than just using that part which corresponds to photon energies greater than the band gap. This procedure could give rise to a small error, because the specimens are partially transparent for some of this wavelength range; however, the error in the derived temperatures is not expected to exceed 10 °C. Figure 7 shows some of graphs of the function $f_e(\lambda)$ for the doped samples. These show that the specimens are

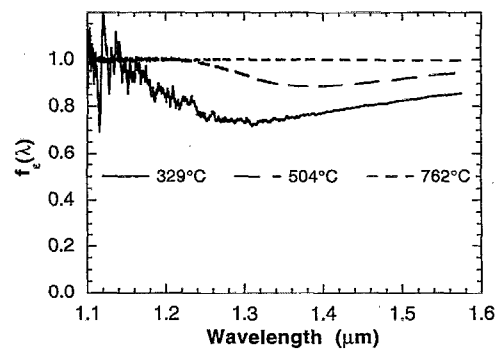


FIG. 7. The normalized spectral emissivity $f_e(\lambda)$ of heavily doped silicon for a range of temperatures.

partially transparent at temperatures below 780 °C, for wavelengths just beyond the absorption edge. Figure 8 shows the absorption spectra derived from the spectral emissivities at a number of temperatures, together with theoretical predictions from the model described above. The agreement between these predictions and the experimental results is not as good as it is for the undoped samples. This is probably because the model used to calculate α_{FC} is a very simple one, which assumes the classical λ^2 dependence for free-carrier absorption, and does not attempt to include the details of the scattering mechanisms involved in the process.²⁸ Other processes that are known to affect optical absorption in doped semiconductors include band-gap narrowing and optical transitions between energy bands and impurity bands.^{29,30} Although there have been extensive studies of free-carrier absorption in semiconductors, very few of these have examined the behavior at high temperatures. As a result it is, at present, not appropriate to construct a general model which would apply across the range of wavelengths, temperatures, and doping concentrations involved in this study.

B. Reflection spectra

The reflection spectra of the silicon samples were also recorded for a range of heating power densities. The study

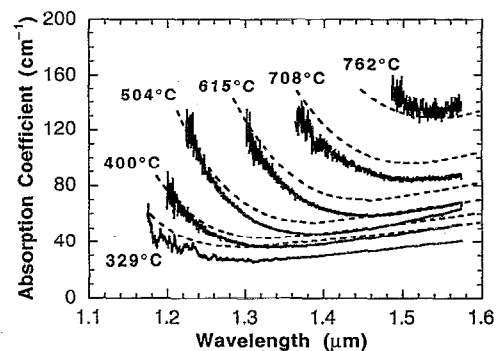


FIG. 8. The absorption spectra of heavily doped silicon at a range of temperatures. The solid curves are experimental results. The dashed curves are theoretical predictions from the model presented in this work.

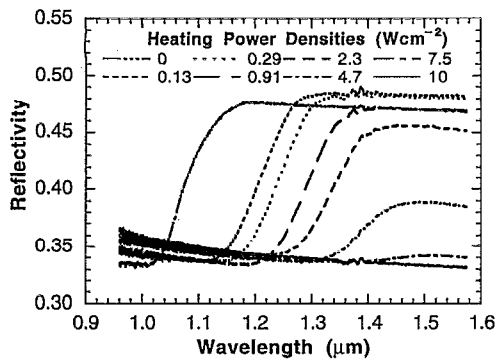


FIG. 9. Reflection spectra for the lightly doped silicon specimen heated with a range of electron-beam power densities.

of reflection spectra is complementary to that of emission spectra, and it is simpler, because the light levels are high and do not display the very strong temperature dependence associated with thermally emitted radiation. The reflection spectra are affected by absorption in the wafer, because light is reflected from the back of the wafer as well as from the front when the wafer is transparent, but it is only reflected from the front if the wafer is opaque. As a result, the reflectivity rises abruptly when the wavelength passes the absorption edge and the wafer becomes transparent. This feature of the reflection spectrum can be used to determine the temperature of a wafer if the temperature dependence of the position of the absorption edge is known.¹¹

Figure 9 shows reflection spectra for the lightly doped silicon, for a range of electron-beam heating power densities. The prominent rising edges in the spectra move to longer wavelengths as the heating power is increased, because the energy gap decreases with temperature and hence the absorption edge occurs at lower photon energies. The effect of the increase in the free-carrier concentration with temperature is also evident in the way in which the magnitude of this rise in reflectivity decreases as the heating power rises until, at 10 W/cm², the rise is indiscernible, because the sample is opaque over the entire wavelength range. For this case, the change in reflectivity with wavelength arises entirely because of the change in the reflectivity of the front surface of the wafer, which is dominated by the wavelength dependence of the refractive index n . It can be seen that this produces a rather a small effect, compared to the change associated with the wafer becoming semitransparent. The reflectivity for wavelengths below the absorption edge varies with the electron-beam heating power density, because of the temperature dependence of n .²⁵ Once again, this is a small effect, and the analysis that follows will ignore both the wavelength and temperature dependence of n . The small oscillations in the reflectivity seen around 1.38 μm are an experimental artifact, which is thought to result from the absorption of the radiation by water vapor in the air. Although most of this effect was removed by the calibration procedure, it seems that small fluctuations in the humidity occurred, producing these artifacts.

The reflectivity of the sample, for the p and s polarizations, is given by

$$R_{p(s)}^* = R_{p(s)} \left(1 + \frac{(1 - R_{p(s)})^2 \exp(-2\alpha l)}{[1 - R_{p(s)}^2 \exp(-2\alpha l)]} \right), \quad (17)$$

where the symbols have the meanings given before.¹¹ The reflectivity for unpolarized light is given by

$$R_u^* = (R_p^* + R_s^*)/2. \quad (18)$$

In principle, the absorption coefficient could be calculated for that part of the spectrum where the specimen is semitransparent, using expressions (17) and (18), but this procedure is impractical for several reasons. The calculation would be very sensitive to the values used for the reflectivities, which would cause a problem because the accuracy of the reflectivity measurement is only $\sim \pm 5\%$. The reflectivity of the sample at room temperature, for wavelengths just below the absorption edge, is ~ 0.334 according to Fig. 9, whereas a value of 0.319 is expected for silicon at room temperature at a wavelength of 0.99 μm .³¹ One error arises because the absolute value of the reflectivity of the gold film used for calibration is not known, and hence it has been assumed to equal one for the purposes of this study. A second error is introduced by the possibility that the light may be partially polarized by reflection from the mirrors in the optical system, which would affect the reflectivity of the sample. Another problem, which rules out the direct use of expressions (17) and (18), is that the wavelength and temperature dependence of R_p and R_s could not be ignored.

Fortunately, these difficulties do not prevent the reflection spectra from being useful. The problems described above can be avoided if the spectra are scaled using the method described previously, where the change in reflectivity at the absorption edge is described in terms of the quantity f_R , which is given by

$$f_R = (R_u^* - R_{u,\alpha=\infty}^*) / (R_{u,\alpha=0}^* - R_{u,\alpha=\infty}^*), \quad (19)$$

where $R_{u,\alpha=\infty}^*$ is the reflectivity for unpolarized light when the sample is opaque, and $R_{u,\alpha=0}^*$ is that when the sample is completely transparent.¹¹ The use of R_u^* , rather than a separate term for each polarization, is justified for the reasons given in Sec. III A above. Once f_R has been calculated, the absorption coefficient can be obtained from the expression¹¹

$$\alpha = \frac{1}{2l} \ln \left(\frac{1 - (1 - f_R) R_u^2}{f_R} \right). \quad (20)$$

Although this expression contains R_u , it is not very sensitive to the value used, and R_u is fixed at 0.32 throughout this study.

The use of this technique for obtaining the absorption spectrum was tested by examining the case where the sample is at room temperature. It was assumed that $R_{u,\alpha=0}^*$ corresponds to the reflectivity beyond the absorption edge, whereas $R_{u,\alpha=\infty}^*$ corresponds to the reflectivity at a wavelength just before the edge. Figure 10 compares the absorption spectrum obtained using this method to recent mea-

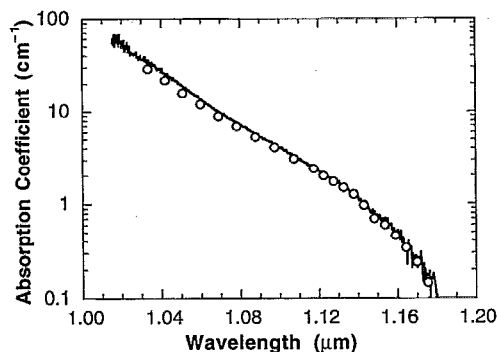


FIG. 10. The absorption spectrum of the lightly doped silicon at room temperature. The solid line is the spectrum calculated from the reflection spectrum shown in Fig. 9. The points are taken from a recent review of the optical properties of silicon (see Ref. 31).

surements, published in a review of the optical properties of silicon.³¹ The excellent agreement between the two sets of data suggests that the absorption spectrum can be obtained from f_R . However, at higher temperatures the silicon does not become completely transparent beyond the absorption edge, and as a result there is some difficulty in determining a value for $R_{u,\alpha=0}^*$. In the analysis that follows the value used for $R_{u,\alpha=0}^*$ is that which corresponds to the reflectivity just beyond the absorption edge, when the sample was heated with a power density of 0.13 W/cm², which was 0.486. This value was used because this was the lowest power density used in this set of experiments, and hence α was very small beyond the absorption edge. The room-temperature value was not used, because the temperature dependence of n would introduce some error. In practice, the method is not very sensitive to the value used for $R_{u,\alpha=0}^*$ provided that the derived values of α are greater than 1 cm⁻¹.

Figure 11 shows the absorption spectra derived by using this method to interpret the spectra shown in Fig. 9. For the heating power densities of 0.91, 2.3, 4.7, and 7.5 W/cm², the results are compared with those shown in Fig. 5 for temperatures of 432, 501, 593, and 687 °C, which were obtained using the same set of heating power densities. Figure 11 shows that the absorption spectra derived from reflection spectra are very similar to those obtained from emission spectra. The result for heating with 0.29 W/cm² is compared to the absorption spectrum at 343 °C, which was obtained in an experiment where the heating power density was 0.26 W/cm². Again, the results are very similar, because the temperatures should not differ by more than 15 °C. The advantage of reflection experiments is that they can be conducted at any temperature, as has been shown with the room-temperature measurements. For the heating power density of 0.13 W/cm², the emission spectrum was too weak to measure, but a value for the temperature of the sample can be obtained by using the model for α , described in Sec. III A, to fit the results. The fit shown in Fig. 11 indicates that the temperature of the sample was 282 °C.

Figure 12 shows some reflection spectra from the

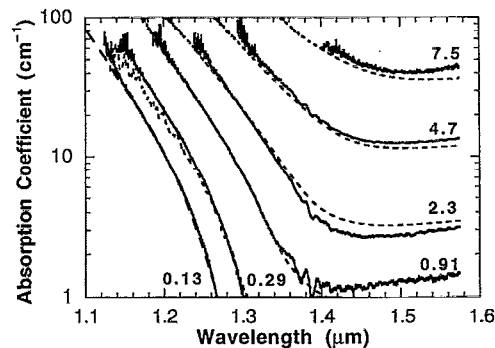


FIG. 11. The solid curves are the absorption spectra of lightly doped silicon at a range of electron-beam heating power densities, as calculated from the reflection spectra shown in Fig. 9. The power densities corresponding to each curve are indicated by a number, which is in units of W/cm². The dashed curves next to the spectra for 0.91, 2.3, 4.7, and 7.5 W/cm², are the absorption spectra, shown in Fig. 5, for temperatures of 432, 501, 593, and 687 °C, respectively. These dashed curves were obtained using the same heating power densities, but were calculated from the emission spectra. The dashed curve next to the spectrum for 0.29 W/cm² is the absorption spectrum at 343 °C, which was obtained from an experiment where the heating power density was 0.26 W/cm². The dashed curve next to the spectrum for 0.13 W/cm² is a fit using the theoretical model, which implies that the temperature was 282 °C.

heavily doped silicon specimen. The rise in reflectivity at the absorption edge is smaller than that for the lightly doped samples because of the effect of free-carrier absorption, which is significant, even at room temperature. However, the shift in the position of the absorption edge with temperature is still evident as the electron-beam power density is increased. Since values for $R_{u,\alpha=0}^*$ cannot be obtained from these experiments, the results were interpreted by using the values used in the analysis of the reflection spectra of the lightly doped silicon. Since n is not a strong function of the doping over the wavelength range of interest here, the error introduced by this approximation is negligible.¹⁴

The resulting absorption spectra are shown in Fig. 13, together with predictions from the theory presented above. For the heating power densities of 1.1 and 2.8 W/cm², the temperatures used for the theoretical predictions were those determined from the analysis of emission spectra ob-

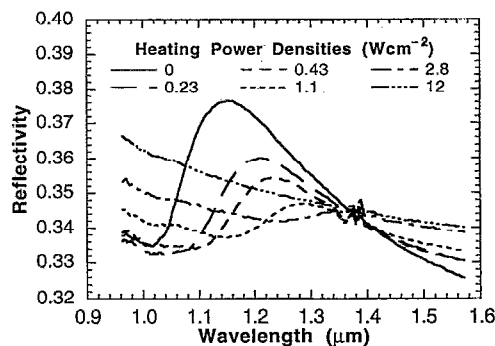


FIG. 12. Reflection spectra for the heavily doped silicon specimen heated with a range of electron-beam power densities.

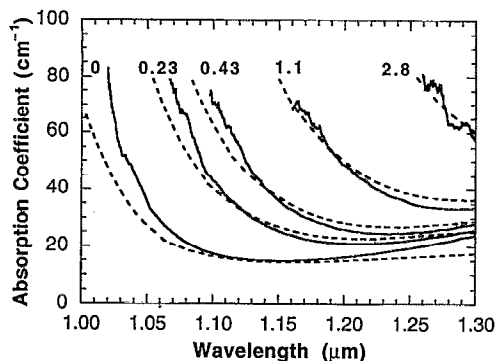


FIG. 13. The solid curves are the absorption spectra of heavily doped silicon at a range of electron-beam heating power densities, as calculated from the reflection spectra shown in Fig. 12. The power densities corresponding to each curve are indicated by a number, which is in units of W/cm^2 . The dashed curves next to the spectra are theoretical predictions. The temperatures used to make these predictions were 22, 155, 212, 329, and 481°C for the power densities of 0, 0.23, 0.43, 1.1, and $2.8 \text{ W}/\text{cm}^2$, respectively.

tained at these power densities, which were 329 and 481°C respectively. For the power densities of 0.23 and $0.43 \text{ W}/\text{cm}^2$, the temperatures were calculated to be 155 and 212°C , respectively, using the data discussed in Sec. III C. The absorption spectra were only calculated for wavelengths near the absorption edge, because the reflectivity change is smaller than it is in the case of the lightly doped silicon, which causes the errors to be larger. In particular, the effects of the wavelength dependence of n introduce significant errors. The experimental artifacts resulting from absorption of the radiation in the waveband between 1.35 and $1.43 \mu\text{m}$ also introduce large errors, so the upper limit of the spectra in Fig. 13 was set at $1.3 \mu\text{m}$. The experimental results broadly follow the trends predicted by the theory, but there are significant differences in the details. In particular, the measured absorption edges, at temperatures below $\sim 200^\circ\text{C}$, occur at wavelengths significantly greater than those predicted by the theory. This “red shift” of the absorption edge in heavily doped silicon is consistent with previous studies and is believed to result from the Coulombic interaction of carriers and impurities.³² A high concentration of impurities not only affects free-carrier absorption but also alters the mechanisms of interband transitions.^{29,30}

C. The temperature dependence of the total emissivity of silicon

In general, in a rapid thermal processing system in which a wafer is thermally isolated from its surroundings, its steady-state temperature is governed by the equation

$$\eta P = 2\sigma[\epsilon_{\text{total}}(T)T^4 - a_{\text{total}}(T)T_0^4], \quad (21)$$

where T is the absolute temperature, $\epsilon_{\text{total}}(T)$ is the total hemispherical emissivity, and $a_{\text{total}}(T)$ is the total hemispherical absorptivity.¹ P is the incident power density, η is the energy coupling efficiency, T_0 is the background temperature, and σ is the Stefan-Boltzmann constant. In the electron-beam heating system, the quantity P can be cal-

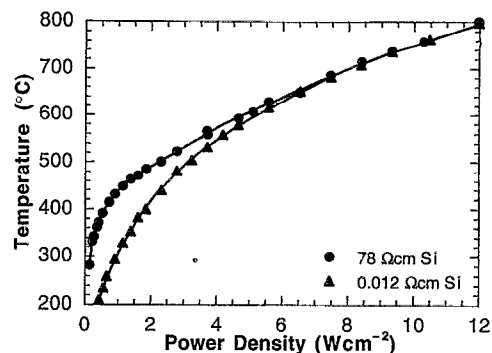


FIG. 14. The temperature of the lightly ($78 \Omega \text{ cm}$) and heavily ($0.012 \Omega \text{ cm}$) doped silicon specimens as a function of heating power density.

culated from $P = VI/A$ where V is the electron beam energy, I is the beam current, and A is the scan area, all of which can be measured. For silicon, η is 0.88 when $V = 20 \text{ kV}$, as was the case in this work.³³ There is some difficulty in deciding what the appropriate values for $a_{\text{total}}(T)$ would be, because this depends on the nature of the background radiation in the apparatus, as well as the sample properties. In general, $a_{\text{total}}(T) \neq \epsilon_{\text{total}}(T)$, unless an object is a grey-body and the incident radiation is independent of angle. This is because the spectrum of the radiation incident on the sample from its surroundings is different from that emitted by a blackbody at the temperature of the sample. However, since the surroundings of the sample remain cool, with $T_0 \sim 300 \text{ K}$, $T_0^4 \ll T^4$ for most of the sample temperatures measured in this work, so that the details of this term are unimportant under most circumstances. For the purposes of this study the approximation that $a_{\text{total}}(T) = \epsilon_{\text{total}}(T)$ will be used. This results in the expression¹¹

$$\epsilon_{\text{total}}(T) = \eta P / [2\sigma(T^4 - T_0^4)]. \quad (22)$$

The error introduced by the use of this approximation is discussed in more detail below.

Figure 14 shows the temperature reached for a given heating power density for the cases of the lightly and heavily doped samples. The temperatures were deduced from emission spectra, except for the point at 282°C for the lightly doped silicon, which was deduced from the reflection spectrum. Above 640°C the results are identical for the two types of specimen, but below this temperature the curves diverge, with the lightly doped sample showing a temperature rise which is almost twice that of the heavily doped specimen when the power density is $0.45 \text{ W}/\text{cm}^2$. The total hemispherical emissivities, calculated using expression (22), are shown as functions of temperature in Fig. 15. The total emissivity of the heavily doped silicon is approximately constant at a value of ~ 0.7 , whereas the lightly doped silicon has an emissivity of 0.12 at 280°C , which rises to a limiting value of 0.7 at $\sim 650^\circ\text{C}$. At present there are no published data for the total hemispherical emissivity of silicon to which the current results could be compared directly. However, the behavior of the total emissivity can be understood in terms of the temperature

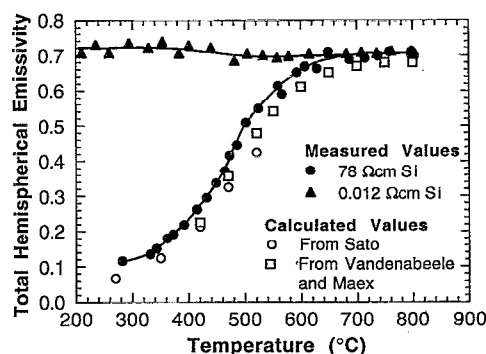


FIG. 15. The total hemispherical emissivity of the lightly (78 Ω cm) and heavily (0.012 Ω cm) doped silicon specimens as a function of temperature. The solid curves are fits to the data points. The open circles are values calculated from the data of Sato (see Ref. 14) and the open squares are values calculated using the data of Vandennebeele and Maex (see Ref. 12).

dependence of the spectral emissivity as observed by Sato.¹⁴ Over the temperature range covered in these experiments the total emissivity is most strongly affected by the behavior of the spectral emissivity in the wavelength range from 2 to 20 μ m. In this wave band optical absorption takes place by interaction of photons with free carriers or with the silicon lattice vibrations. At low temperatures the concentration of free carriers in the silicon is determined by the doping of the silicon, and it is very low in the lightly doped material. As a result, there is little free-carrier absorption in the lightly doped silicon, and the total emissivity is determined by absorption associated with lattice vibrations, which is quite weak and results in a low total emissivity. As the temperature rises, the free-carrier concentration increases, and the accompanying increase in the absorption results in the rapid rise in the total emissivity. In the heavily doped silicon the carrier concentration is high even at room temperature, and the additional thermal generation of carriers produces an insignificant change in the total emissivity.

In order to carry out a more detailed study of these effects a theoretical prediction of the temperature dependence of the total emissivity of silicon was calculated using the spectral data reported by Sato, for a phosphorus-doped sample which was 1.77 mm thick and had a resistivity of 15 Ω cm.¹⁴ This resistivity is high enough for the doping to have a negligible effect on the optical absorption, over the temperature range covered in this work. However, the thickness of the sample was considerably larger than that of the samples used in this work, and this difference must be taken into account. This can be done by calculating the absorption spectrum from the spectral emissivity of the 1.77-mm-thick sample, and then calculating the spectral emissivity for a sample that is 390 μ m thick. The absorption spectra were calculated using the expression

$$\alpha = \frac{1}{d} \ln \left(1 + \frac{\epsilon_{\infty}(\lambda) \epsilon_d(\lambda)}{[\epsilon_{\infty}(\lambda) - \epsilon_d(\lambda)]} \right), \quad (23)$$

where $\epsilon_{\infty}(\lambda)$ is the spectral emissivity for a specimen which is infinitely thick and $\epsilon_d(\lambda)$ is the spectral emissivity for a specimen which is of thickness d .¹⁴ $\epsilon_{\infty}(\lambda)$ was taken to be equivalent to the spectral emissivity of the 1.77-mm-thick specimen at 600 $^{\circ}$ C, at which temperature it was opaque over the whole spectral range studied. This is an approximation because it does not consider the temperature dependence of the reflectivity of the silicon surface. From Sato's data it was possible to calculate α for the range of wavelengths where $\alpha < 20$ cm⁻¹. This is not enough information to calculate $\epsilon_d(\lambda)$ for a sample that is 390 μ m thick, because this specimen does not become effectively opaque until α exceeds 100 cm⁻¹. In order to obtain an estimate of α for wavelengths where $\alpha > 20$ cm⁻¹ the data was extrapolated in a manner consistent with Sato's observation that the free-carrier absorption was proportional to λ^2 , for $\lambda > 5$ μ m. It was found that Sato's values for α could be approximated by the expression

$$\alpha = 0.175 \lambda^2 T^{3/2} \exp(-7000/T), \quad (24)$$

where T is in K, λ is in μ m, and α is in cm⁻¹. The exponential term is included on the basis that the free-carrier absorption is expected to be proportional to the carrier concentration, which is approximately proportional to $T^{3/2} \exp(-7000/T)$.¹² Expression (24) does not attempt to take account of any of the details of the free-carrier absorption process, and merely serves as an extrapolation of the available data. Because Sato's data only covered wavelengths up to 15 μ m, expression (24) was also used to extrapolate the absorption spectra to longer wavelengths.

Using these methods, $\epsilon_d(\lambda)$ was calculated for a 390- μ m-thick sample, at temperatures of 270, 350, 420, 470, and 520 $^{\circ}$ C. The calculated spectral emissivities were then multiplied by the emission spectra for a blackbody at these temperatures, and the resulting products were integrated to obtain values for the total emissivity. It should be noted that Sato's data seem to refer to the normal emissivity, so the integration gives the total normal emissivity, rather than the total hemispherical emissivity. The difference between the two should be rather small for a material with the optical properties of silicon, and is ignored in these calculations, bearing in mind the various other approximations involved.

Similar calculations were performed using the expression given by Vandennebeele and Maex, who found that α could be described by

$$\alpha = 4.15 \times 10^{-5} \lambda^{1.51} T^{2.95} \exp(-7000/T), \quad (25)$$

for $1.5 < \lambda < 5$ μ m,

where T is in K, λ is in μ m, and α is in cm⁻¹.¹² In these calculations it was assumed that expression (25) could be used over the entire wavelength range of interest, rather than just the wavelength range specified. Using this approach, the total emissivity was calculated for temperatures between 420 and 800 $^{\circ}$ C.

The calculated values of the total emissivity are shown in Fig. 15. Although there is some difference between the predicted and measured values, their temperature depen-

dences display the same trend. The measured value at 280 °C is ~ 0.12 , whereas the value predicted from Sato's data is ~ 0.07 . This difference arises because the extrapolation used to extend Sato's results to wavelengths above 15 μm underestimates the absorption at low temperatures, because it takes no account of the absorption due to lattice vibrations, which is significant. The values calculated using Eq. (25) show good agreement with the results over the temperature range from 400 to 800 °C. Expression (25) was not used for temperatures below 400 °C, because it does not predict the lattice absorption, which dominates the absorption spectrum of lightly doped silicon at temperatures below 400 °C.

The errors in the measured values of the total emissivity are expected to be $\sim 8\%$, if one takes into account typical errors in the measurement of the electron-beam heating parameters, which are $\sim \pm 6\%$, a temperature measurement error of ± 5 °C, and the error in η which is unlikely to exceed $\pm 3\%$. Another source of error would arise if there was significant heat lost from the edges of the sample, as compared to that lost from the surfaces. This would result in a slight overestimate of the total emissivity. For a 10 mm square sample that is 390 μm thick, the area of the edges is $< 8\%$ of the area of the surfaces. Furthermore, the sides of the specimen carrier tend to reflect heat back onto the chip, in the mounting arrangement used in this work. Hence, the error from edge losses is expected to be less than 5%.

The errors which arise from the approximation that $a_{\text{total}}(T) = \epsilon_{\text{total}}(T)$, and from error in the value of T_0 become increasingly significant as the temperature decreases. It is worth examining the role of the background radiation in determining the specimen temperature in some detail. If it were true to say that $a_{\text{total}}(T) = \epsilon_{\text{total}}(T)$, then the background radiation could be safely ignored for temperatures above ~ 400 °C, because under these circumstances $T^4 \gg T_0^4$. However, if $\epsilon_{\text{total}}(T) \ll a_{\text{total}}(T)$, then it is possible that the absorbed radiation might not be insignificant compared to the emitted radiation. When the silicon is semi-transparent in the infrared, one expects that $a_{\text{total}}(T) > \epsilon_{\text{total}}(T)$, if $T > T_0$, because the absorption coefficient rises with the wavelength, and the spectrum of the background radiation lies deeper in the infrared than the emission spectrum of a blackbody at the specimen temperature. However, calculations based on expression (24) suggest that $a_{\text{total}}(T)$ is at most $\sim 2\epsilon_{\text{total}}(T)$. This implies that the background radiation can be ignored for temperatures greater than ~ 500 °C. For temperatures below this, the use of expression (22) results in a slight underestimate of the total emissivity of the undoped silicon, but the magnitude of the error should not exceed 8%. For the heavily doped silicon, free-carrier absorption is large throughout the infrared, and the spectral emissivity varies slowly over the wavelength range from 1 to 15 μm .¹⁴ As a result, it is expected that $a_{\text{total}}(T) \approx \epsilon_{\text{total}}(T)$, for the whole range of temperatures covered here, and the error from using expression (22) should be negligible.

IV. CONCLUSIONS

Studies of the spectral emissivity of the lightly doped silicon permitted the measurement of the absorption coefficient over the wavelength range from 1.1 to 1.6 μm and the temperature range from 330 to 800 °C. The data showed good agreement with a model which is a development of the approaches suggested by Macfarlane *et al.*¹⁵ and Sturm and Reaves.²⁰ It was found that it is necessary to take account of all the phonon-assisted absorption processes identified by McLean in order to obtain satisfactory agreement between the theory and experimental results.¹⁶ Further improvements might be achieved by performing detailed calculations of the absorption mechanisms involving two phonons, and by including the relaxation processes which cause thermal broadening of features in the absorption spectra. One difficulty which stands in the way of these improvements is the development of a very accurate model for the free-carrier absorption, which will require a more detailed study of the absorption for photon energies below the band gap. This might also improve the agreement between the theory and experiment for the case of heavily doped silicon.

It was shown that reflection spectra can be used to determine the absorption spectrum, for the range where $1 \text{ cm}^{-1} < \alpha < 70 \text{ cm}^{-1}$. By fitting the spectrum to the theoretical model for α , it is possible to deduce the temperature of the lightly doped silicon, which is especially useful for temperatures less than 300 °C, where the thermal emission is very weak.

The total hemispherical emissivity of the lightly doped silicon was found to be a strong function of the temperature. The large increase in the total emissivity as the temperature rises is a consequence of the increasing free-carrier concentration. The results were consistent with values predicted by integrating Sato's data for the temperature dependence of the spectral emissivity and with predictions based on the data of Vandenabeele and Maex. For heavily doped specimens the total emissivity remained approximately constant between 200 and 800 °C. The error in the measured values of the total emissivity is $\sim 10\%$, and contains both systematic and random components, with the random element dominating over most of the temperature range covered in this study. For greater accuracy, larger or thinner samples could be used, in order to make the edge losses insignificant, and the chamber and stage could be cooled so that the background radiation term would be minimized.

ACKNOWLEDGMENTS

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